Practice Exercises: Teaching Block 4

- Numbers
- Exception Handling
- File I/O
- Fill in the Gaps

This set of exercises is in addition to those included directly in lecture slides (and extra reading materials), which you should also attempt.

Question 1

- Are recursion and iteration in Java the same? Write a code to generate Fibonacci numbers using a recursive method.

Question 2

- Write the Java code that generates n random numbers, then sorts them in descending order using bubble sort. Print the original number sequence as well as the sorted sequence

Question 3

- Given the following code explain:
  1) What the code does
  2) What will happen if the file doesnt exist
  3) What will happen if num2 = 0

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;

public class NumberDivider {
      public static void main(String[] args) {
      try {
                        // Read numbers from the text file
                    BufferedReader reader = new BufferedReader(new
FileReader("numbers.txt"));
                        String line1 = reader.readLine();
                        String line2 = reader.readLine();
                        reader.close();
```

Question 3

```
    // Convert the strings to numbers
            double num1 = Double.parseDouble(line1);
            double num2 = Double.parseDouble(line2);

            // Divide the numbers
            double result = divide(num1, num2);
            System.out.println("Result: " + result);
            } catch (IOException e) {
                 System.out.println("Error reading the file: " + e.getMessage());
            } catch (NumberFormatException e) {
                 System.out.println("Invalid number format in the file: " +
e.getMessage());
            } catch (ArithmeticException e) {
                 System.out.println("Error: " + e.getMessage());
        }
    }

    private static double divide(double num1, double num2) throws ArithmeticExcepti
{
        if (num2 == 0) {
            throw new ArithmeticException("Division by zero");
        }
        return num1 / num2;
    }
}
```